



LEVEL-II (C.W)

INTEGRATION BY THE METHOD OF
SUBSTITUTION

1. $\int \{1 + 2 \tan x (\tan x + \sec x)\}^{\frac{1}{2}} dx =$
 (where $x \in \left(0, \frac{\pi}{2}\right)$)

1) $\log |\sec x (\sec x - \tan x)| + c$

2) $\log |\sec x (\sec x + \tan x)| + c$

3) $\log \left| \frac{\sec x}{\sec x + \tan x} \right| + c$

4) $\log |\cos x (\sec x + \tan x)| + c$

2. $\int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2 \log x]}{x^4} dx$ is equal to

1) $\frac{(x^2+1)^{3/2}}{x^3} \left[\frac{2}{3} - \log \left(\frac{x^2+1}{x^2} \right) \right] + c$

2) $\frac{(x^2+1)^{3/2}}{3x^3} \left[\log \left(\frac{x^2+1}{x^2} \right) - \frac{2}{3} \right] + c$

3) $\frac{2(x^2+1)^{3/2}}{3x^3} \left[\frac{1}{3} - \frac{1}{2} \log \left(\frac{x^2+1}{x^2} \right) \right] + c$

4) $\frac{(x^2+1)^{3/2}}{3x^3} \left[\log \left(\frac{x^2+1}{x^2} \right) + \frac{2}{3} \right] + c$

3. $\int \frac{mx^{m+2n-1} - nx^{n-1}}{x^{2m+2n} + 2x^{m+n} + 1} dx$ is equal to

1) $\frac{x^m}{x^{m+n} + 1} + c$

2) $\frac{x^n}{x^{m+n} + 1} + c$

3) $\frac{x^{m+n} - 1}{x^{m+n} + 1} + c$

4) $-\frac{x^n}{x^{m+n} + 1} + c$

4. $\int \frac{\cos x}{\cos 3x} dx =$

1) $\frac{1}{2\sqrt{3}} \log \left| \frac{1 + \tan x}{1 - \tan x} \right| + c$

2) $\frac{1}{2\sqrt{3}} \log \left| \frac{1 + \sqrt{3} \tan x}{1 - \sqrt{3} \tan x} \right| + c$

3) $\frac{1}{\sqrt{3}} \log \left| \frac{1 + \sqrt{3} \tan x}{1 - \sqrt{3} \tan x} \right| + c$

4) $\frac{2}{\sqrt{3}} \log \left| \frac{1 - \tan x}{1 + \tan x} \right| + c$

5. If $f(x) = \frac{1}{3} \cot^3 x - \cot x + \int \cot^4 x dx$ and

$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $f(x) =$

1) $\pi - x$

2) $x - \pi$

3) $\frac{\pi}{2} - x$

4) x

6. $\int \frac{\sin^8 x - \cos^8 x}{1 - 2 \sin^2 x \cos^2 x} dx =$

1) $\frac{1}{2} \cos 2x + c$

2) $-\frac{1}{2} \cos 2x + c$

3) $\frac{-1}{(1 + \tan x)^2} + c$

4) $\frac{-1}{2} \sin 2x + c$

7. $\int \frac{1}{x\sqrt{1+x^n}} dx = \dots$

1) $\frac{1}{n} \log \sqrt{\frac{1+x^n}{1-x^n}} + c$

2) $\frac{1}{n} \log \left| \frac{\sqrt{1+x^n} - 1}{\sqrt{1+x^n} + 1} \right| + c$

3) $\frac{1}{n} \log \left| \frac{1+x^n}{1-x^n} \right| + c$

4) $\frac{1}{n} \log \left| \frac{\sqrt{1+x^n} + 1}{\sqrt{1+x^n} - 1} \right| + c$

8. $\int (x^x)^x (2x \log_e x + x) dx$ is equal to

1) $x^{(x^x)} + C$

2) $(x^x)^x + C$

3) $x^x \cdot \log_e x + C$

4) $(x^x) + C$

9. $\int \frac{dx}{\sin^{1/2} x \cos^{7/2} x} =$

1) $\frac{2}{5}(5 + \tan^2 x) + c$ 2) $\frac{2}{5}\sqrt{\tan x} + c$

3) $\frac{2}{5}\tan x(5 - \tan^2 x) + c$

4) $\frac{2}{5}\sqrt{\tan x}(5 + \tan^2 x) + c$

10. $\int \frac{1}{x\{6(\log x)^2 + 7\log x + 2\}} dx =$

1) $\log \left| \frac{2\log x + 1}{3\log x + 2} \right| + C$ 2) $\log \left| \frac{2\log x - 1}{3\log x + 2} \right| + C$

3) $\log \left| \frac{2\log x - 1}{3\log x - 2} \right| + C$ 4) $\log \left| \frac{3\log x + 1}{2\log x - 2} \right| + C$

11. $\int \frac{x^{1/2}}{x^{1/2} - x^{1/3}} dx$ is equal to

1) $6 \left[\frac{x}{6} + \frac{x^{5/6}}{5} + \frac{x^{2/3}}{4} + \frac{x^{1/2}}{3} + \frac{x^{1/3}}{2} + x^{1/6} + \log \left(x^{1/6} - 1 \right) \right] + C$

2) $6 \left[\frac{x}{6} - \frac{x^{5/6}}{5} + \frac{x^{2/3}}{4} - \frac{x^{1/2}}{3} + \frac{x^{1/3}}{2} + x^{1/6} - \log \left(x^{1/6} - 1 \right) \right] + C$

3) $6 \left[\frac{x}{6} + \frac{x^{5/6}}{5} - \frac{x^{2/3}}{4} + \frac{x^{1/2}}{3} - \frac{x^{1/3}}{2} + x^{1/6} - \log \left(x^{1/6} - 1 \right) \right] + C$

4) $6 \left[\frac{x}{6} + \frac{x^{5/6}}{5} + \frac{x^{2/3}}{4} + \frac{x^{1/2}}{3} + \frac{x^{1/3}}{2} - x^{1/6} + \log \left(x^{1/6} - 1 \right) \right] + C$

12. $\int \frac{dx}{(\sqrt{1+x^2} - x)^n} (n \neq \pm 1) = \frac{1}{2} \left(\frac{z^{n+1}}{n+1} + \frac{z^{n-1}}{n-1} \right) + C$

where

1) $z = x - \sqrt{1+x^2}$ 2) $z = \sqrt{1+x^2} - x$

3) $z = x + \sqrt{1+x^2}$ 4) $z = x - \sqrt{1-x^2}$

SUBSTITUTION OF TRIGONOMETRIC FUNCTIONS

13. $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to

1) $\frac{2(\sqrt{x}-1)}{\sqrt{1-x}} + c$ 2) $\frac{2(1-\sqrt{x})}{\sqrt{1-x}} + c$

3) $\frac{(\sqrt{x}-1)}{\sqrt{1-x}} + c$ 4) $\frac{(\sqrt{x}+1)}{\sqrt{1-x}} + c$

14. $\int \frac{dx}{x^2 \sqrt{x^2 + a^2}} =$

1) $-\frac{\sqrt{x^2 + a^2}}{a^2 x} + c$ 2) $\frac{\sqrt{x^2 + a^2}}{a^2 x} + c$

3) $-\frac{a^2 x}{\sqrt{x^2 + a^2}} + c$ 4) $\frac{a^2 x}{\sqrt{x^2 + a^2}} + c$

15. Let $f(x) = \int \frac{x^2}{(1+x^2)(1+\sqrt{1+x^2})} dx$ and

$f(0) = 0$... Then $f(1)$ is

1) $\log(1+\sqrt{2})$ 2) $\log(1+\sqrt{2}) + \frac{\pi}{4}$

3) $\log(1+\sqrt{2}) - \frac{\pi}{4}$ 4) $\log(\sqrt{2}-1)$

RATIONAL INDEX

16. $\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1+\sqrt[3]{x})} dx$ is equal to

1) $\frac{3}{2}x^{2/3} + 6 \tan^{-1} x^{1/6} + c$

2) $\frac{3}{2}x^{2/3} - 6 \tan^{-1} x^{1/6} + c$

3) $-\frac{3}{2}x^{2/3} + 6 \tan^{-1} x^{1/6} + c$

4) $-\frac{3}{2}x^{2/3} - 6 \tan^{-1} x^{1/6} + c$

17. $\int x^{1/3} (2+x^{2/3})^{1/4} dx$ is equal to

- 1) $\frac{2}{3} (2+x^{2/3})^{9/4} + \frac{12}{5} (2+x^{2/3})^{5/4} + c$
- 2) $\frac{2}{3} (2+x^{2/3})^{9/4} - \frac{12}{5} (2+x^{2/3})^{5/4} + c$
- 3) $\frac{1}{3} (2+x^{2/3})^{9/4} - \frac{12}{5} (2+x^{2/3})^{5/4} + c$
- 4) $\frac{1}{3} (2+x^{2/3})^{9/4} + \frac{12}{5} (2+x^{2/3})^{5/4} + c$

18. $\int \frac{1}{x^{1/2} (1+x^2)^{5/4}} dx$ is equal to

- 1) $\frac{-2\sqrt{x}}{\sqrt[4]{1+x^2}} + c$
- 2) $\frac{2\sqrt{x}}{\sqrt[4]{1+x^2}} + c$
- 3) $\frac{-\sqrt{x}}{\sqrt[4]{1+x^2}} + c$
- 4) $\frac{\sqrt{x}}{\sqrt[4]{1+x^2}} + c$

19. $\int \frac{dx}{(x-p)\sqrt{(x-p)(x-q)}} =$

- 1) $\frac{2}{p-q} \sqrt{\frac{x-p}{x-q}} + c$
- 2) $\frac{-2}{p-q} \sqrt{\frac{x-q}{x-p}} + c$
- 3) $\frac{1}{\sqrt{(x-p)(x-q)}} + c$
- 4) $\sqrt{\frac{x-q}{x-p}} + c$

20. $\int \frac{dx}{(x-2)^{7/8} (x+3)^{9/8}} =$

- 1) $\frac{8}{5} \left(\frac{x-2}{x+3} \right)^{1/8} + c$
- 2) $\frac{8}{3} \left(\frac{x-2}{x+3} \right)^{1/8} + c$
- 3) $\frac{8}{5} \left(\frac{x+2}{x+3} \right)^{1/8} + c$
- 4) $\frac{3}{8} \left(\frac{x+2}{x+3} \right)^{1/8} + c$

INTEGRATION BY PARTS

I. Problems based on $\int uv dx$

21. $\int \frac{\log(1-x)}{x^2} dx$ equal to

- 1) $\left[\left(1 + \frac{1}{x} \right) \log(1-x) - \log x \right] + c$
- 2) $\left(1 - \frac{1}{x} \right) \log(1-x) - \log x + c$
- 3) $\left(\frac{1}{x} - 1 \right) \log(1-x) + \log x + c$
- 4) $\left(\frac{1}{x} - 1 \right) \log(1-x) - \log x + c$

22. $\int \{f(x)g''(x) - f''(x)g(x)\} dx =$

- 1) $f(x)g'(x) - g(x)f'(x) + c$
- 2) $g(x)f'(x) - f(x)g'(x) + c$
- 3) $g(x)f'(x) + f(x)g'(x) + c$
- 4) $f(x)g(x) + c$

23. If $\int f(x)dx = g(x)$ then $\int x^3 f(x^2)dx =$

- 1) $\frac{1}{2} \{x^2 g(x^2) - \int g(x^2)dx^2\}$
- 2) $\frac{1}{2} \{x^2 g(x^2) - \int g(x^2)dx\}$
- 3) $\frac{1}{2} \{x^2 g(x^2) + \int g(x^2)dx^2\}$
- 4) $\frac{1}{2} \{x^2 g(x^2) + \int g(x^2)dx\}$

24. If $\int f(x)dx = g(x) + c$, then $\int f^{-1}(x)dx$ is equal to

- 1) $xf^{-1}(x) + c$
- 2) $f[g^{-1}(x)] + c$
- 3) $xf^{-1}(x) - g[f^{-1}(x)] + c$
- 4) $g^{-1}(x) + c$

25. $\int \cos x \log \left(\tan \frac{x}{2} \right) dx =$

- 1) $\sin x \log |\tan x| - x + c$
- 2) $-\sin x \log \left| \tan \frac{x}{2} \right| + x + c$
- 3) $-\sin x \log \left| \tan \frac{x}{2} \right| - x + c$
- 4) $\sin x \log \left| \tan \frac{x}{2} \right| - x + c$

26. $\int \frac{x^2(x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx = \frac{-x^2}{x \tan x + 1} + f(x) + c$

then $f(x) =$

- 1) $\log|x \sin x + \cos x| + c$
- 2) $\log|x \cos x + \sin x| + c$
- 3) $2 \log|x \sin x + \cos x| + c$
- 4) $2 \log|x \cos x + \sin x| + c$

27. $\int x^3 (\log x)^2 dx =$

- 1) $\frac{x^4}{4} \left[(\log x)^2 - \frac{\log x}{2} + \frac{1}{8} \right] + c$
- 2) $\frac{x^4}{4} \left[(\log x)^2 + \frac{\log x}{2} + \frac{1}{8} \right] + c$
- 3) $(\log x)^2 + 2 \log x + 8 + c$
- 4) $\frac{x^4}{4} \left[(\log x)^2 - \frac{\log x}{2} - \frac{1}{8} \right] + c$

II. Problems based on

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + c;$$

28. If $\int e^x (\sin^2 2x - 8 \cos 4x) dx = e^x f(x) + c$,

then $f(x) =$

- 1) $\sin^2 2x - \sin 4x$
- 2) $\sin^2 2x + 4 \cos 4x$
- 3) $\sin^2 2x + 2 \cos 4x$
- 4) $\sin^2 2x - 2 \sin 4x$

29. $\int \left(\frac{\cos x - 1}{\sin x + 1} \right) e^x dx$ is equal to

- 1) $\frac{e^x \cos x}{1 + \sin x} + c$
- 2) $c - \frac{e^x \sin x}{1 + \sin x}$
- 3) $c - \frac{e^x}{1 + \sin x}$
- 4) $c - \frac{e^x \cos x}{1 + \sin x}$

30. The antiderivative of

$$f(x) = \log(\log x) + (\log x)^{-2}$$

whose graph passes through (e, e) is

- 1) $x \left[\log(\log x) + (\log x)^{-1} \right]$
- 2) $x \left[-\log(\log x) + (\log x)^{-1} \right] + e$
- 3) $x \left[\log(\log x) - (\log x)^{-1} \right] + 2e$
- 4) $x \left[\log(\log x) - (\log x)^{-1} \right] + 3e$

31. $\int \left[\frac{1 + \sin(\log x)}{1 + \cos(\log x)} \right] dx =$

- 1) $\frac{x}{1 + \cos(\log x)} + c$
- 2) $x \tan \frac{\log x}{2} + c$
- 3) $-x \cot \frac{\log x}{2} + c$
- 4) $\frac{x}{1 + \sin(\log x)} + c$

32. $\int x^3 \left[\frac{1 - \sin(4 \log x)}{1 - \cos(4 \log x)} \right] dx =$

- 1) $x^4 \cot(2 \log x) + c$
- 2) $e^{4x} \frac{1}{1 - \cos(4 \log x)} + c$
- 3) $\frac{-x^4}{4} \cot(2 \log x) + c$
- 4) $\frac{x^4}{4} \cot(2 \log x) + c$

33. If $x \in \left(0, \frac{\pi}{2} \right)$ then $\int e^{\frac{-x}{2}} \frac{\sqrt{1 - \sin x}}{1 + \cos x} dx =$

- 1) $e^{-x/2} \sec \frac{x}{2} + c$
- 2) $-e^{\frac{-x}{2}} \sec \frac{x}{2} + c$
- 3) $e^{\frac{x}{2}} \sec \frac{x}{2} + c$
- 4) $-e^{\frac{x}{2}} \sec \frac{x}{2} + c$

34. The value of $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1 - x^n) \sqrt{1 - x^{2n}}} dx$ is

- 1) $e^x \frac{\sqrt{1 - x^n}}{1 - x^n} + c$
- 2) $e^x \frac{\sqrt{1 + x^{2n}}}{1 - x^{2n}} + c$
- 3) $e^x \frac{\sqrt{1 - x^{2n}}}{1 - x^{2n}} + c$
- 4) $e^x \frac{\sqrt{1 - x^{2n}}}{1 - x^n} + c$

35. Let $f(x) = \int e^x (x-1)(x-2)dx$. Then f decreases in the interval

- 1) $(-\infty, -2)$ 2) $(-2, -1)$
3) $(1, 2)$ 4) $(2, +\infty)$

III. Problems based on $\int e^{ax} \sin bx \, dx$ and

$$\int e^{ax} \cos bx \, dx$$

36. $\int e^{2x} (\sqrt{3} \cos x - \sin x) dx =$

- 1) $\frac{e^{2x}}{5} [(2\sqrt{3}+1)\cos x - (\sqrt{3}-2)\sin x] + c$
2) $\frac{e^{2x}}{5} [(2\sqrt{3}+1)\sin x + (\sqrt{3}-2)\cos x] + c$
3) $\frac{e^{2x}}{5} [(2\sqrt{3}+1)\sin x - (\sqrt{3}-2)\cos x] + c$
4) $\frac{e^{2x}}{5} [(2\sqrt{3}+1)\cos x + (\sqrt{3}-2)\sin x] + c$

37. $\int x e^{-5x^2} \sin 4x^2 \, dx = K e^{-5x^2} (A \sin 4x^2 + B \cos 4x^2) + c$
then (K, A, B) (Where $K > 0$) =

- 1) $\frac{1}{82}, 5, 4$ 2) $\frac{1}{82}, -5, 4$
3) $\frac{1}{82}, -5, -4$ 4) $\frac{1}{82}, \frac{-1}{5}, \frac{-1}{4}$

APPLICATIONS OF STANDARD INTEGRALS

38. $\int \frac{1-x^2}{(1+x^2)\sqrt{1+x^4}} dx$ is equal to

- 1) $\sqrt{2} \sin^{-1} \left\{ \frac{\sqrt{2}x}{x^2+1} \right\} + c$
2) $\frac{1}{\sqrt{2}} \sin^{-1} \left\{ \frac{\sqrt{2}x}{x^2+1} \right\} + c$
3) $\frac{1}{2} \sin^{-1} \left\{ \frac{\sqrt{2}x}{x^2+1} \right\} + c$
4) $\frac{1}{\sqrt{2}} \sin^{-1} \left\{ \frac{x^2+1}{\sqrt{2}x} \right\} + c$

39. $\int \frac{\sin \theta - \cos \theta}{\sqrt{\sin 2\theta}} d\theta =$

- 1) $-\log(\sin \theta + \cos \theta + \sqrt{\sin 2\theta}) + c$
2) $\log(\sin \theta + \cos \theta + \sqrt{\sin 2\theta}) + c$
3) $\log(\sin \theta - \cos \theta - \sqrt{\sin 2\theta}) + c$
4) $\log(\tan \theta + \cot \theta) + c$

INTEGRATION OF RATIONAL FUNCTIONS USING PARTIAL FRACTIONS

40. If $\int \frac{1}{a^4 - x^4} dx = A \tan^{-1} \left(\frac{x}{a} \right) + B \log \left| \frac{a+x}{a-x} \right| + C$

then $(A, B) =$

- 1) $\left(-\frac{1}{2a^3}, \frac{1}{4a^3} \right)$ 2) $\left(\frac{1}{2a^3}, \frac{1}{4a^3} \right)$
3) $\left(\frac{1}{2a^3}, \frac{1}{4a} \right)$ 4) $\left(\frac{1}{a}, \frac{1}{2a} \right)$

41. $\int \frac{x^3 - 1}{x^3 + x} dx$ equal to

- 1) $x - \log x + \log(x^2 + 1) - \tan^{-1} x + c$
2) $x - \log x + \frac{1}{2} \log(x^2 + 1) - \tan^{-1} x + c$
3) $x + \log x + \frac{1}{2} \log(x^2 + 1) + \tan^{-1} x + c$
4) $x + \log x - \frac{1}{2} \log(x^2 + 1) - \tan^{-1} x + c$

42. The value of $\int \frac{dx}{x^{13}(x^6-1)}$ is

- 1) $\log \left(\left| \frac{x^6-1}{x^6} \right| \right) + x^{-6} + x^{-12} + c$
2) $\frac{1}{6} \log \left(\left| \frac{x^6-1}{x^6} \right| \right) + \frac{1}{6} x^{-6} + \frac{1}{4} x^{-12} + c$
3) $\frac{1}{6} \left(\log \left| \frac{x^6-1}{x^6} \right| + x^{-6} + \frac{1}{2} x^{-12} \right) + c$
4) $\frac{1}{6} \log \left(\left| \frac{x^6-1}{x^6} \right| \right) + c$

METHODS OF INTEGRATION

43. $\int \frac{dx}{(x+1)\sqrt{x^2+x+1}} =$

- 1) $\text{Cosh}^{-1}\left(\frac{1-x}{\sqrt{3}(1+x)}\right) + c$ 2) $\text{Sin h}^{-1}\left(\frac{1-x}{\sqrt{3}(1+x)}\right) + c$
 3) $-\text{Sinh}^{-1}\left(\frac{1-x}{\sqrt{3}(1+x)}\right) + c$ 4) $-\text{Cosh}^{-1}\left(\frac{1-x}{\sqrt{3}(1+x)}\right) + c$

REDUCTION FORMULAE

44. If $I_{m,n} = \int \frac{x^m}{(\log x)^n} dx$ then

$(m+1)I_{m,n} - nI_{m,n+1} =$

- 1) $\frac{x^m}{(\log x)^n}$ 2) $-\frac{x^m}{(\log x)^n}$
 3) $-\frac{x^{m+1}}{(\log x)^n}$ 4) $\frac{x^{m+1}}{(\log x)^n}$

MISCELLANEOUS PROBLEMS

45. $\int \frac{1}{\sin(x-a)\sin(x-b)} dx =$

- 1) $\frac{1}{\sin(b-a)} \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + c$
 2) $\frac{1}{\sin(a-b)} \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + c$
 3) $\frac{1}{\sin(a+b)} \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + c$
 4) $\frac{1}{\cos(a+b)} \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + c$

46. If $\int \frac{dx}{1+\sin x} = \tan\left(\frac{x}{2} + k\right) + b$, then $k =$

- 1) $-\frac{\pi}{4}$ 2) $\frac{\pi}{2}$
 3) $\frac{5\pi}{4}$ 4) $\frac{\pi}{4}$

47. $\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}} dx$ is equal to

- 1) $\frac{x}{\sqrt{x^4+x^2+1}} + c$ 2) $\frac{\sqrt{x^4+x^2+1}}{x} + c$
 3) $\frac{2x}{\sqrt{x^4+x^2+1}} + c$ 4) $\frac{\sqrt{x^4+x^2+1}}{2x} + c$

48. If $f(x)$ is a polynomial satisfying

$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$, and $f(3) = 82$,

then $\int \frac{f(x)}{x^2+1} dx =$

- 1) $x^3 - x + 2 \tan^{-1} x + c$ 2) $\frac{1}{3}x^3 - x + \tan^{-1} x + c$
 3) $\frac{x^3}{3} - x + 2 \tan^{-1} x + c$ 4) $\frac{1}{3}x^3 + x + 2 \tan^{-1} x + c$

49. If $f(x) = \int \frac{(x^2 + \sin^2 x)}{1+x^2} \sec^2 x dx$ and

$f(0) = 0$ then $f(1) =$

- 1) $1 - \frac{\pi}{4}$ 2) $\frac{\pi}{4} - 1$
 3) $\tan 1 - \frac{\pi}{4}$ 4) $\frac{\pi}{4} - \tan 1$

50. $\int \frac{dx}{\sin x - \cos x + \sqrt{2}}$ is equal to

- 1) $-\frac{1}{\sqrt{2}} \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$ 2) $\frac{1}{\sqrt{2}} \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$
 3) $\frac{1}{\sqrt{2}} \cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$ 4) $-\frac{1}{\sqrt{2}} \cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

51. If $[.]$ denotes greatest integer function and

$x \in \left(\frac{2}{3}, 1\right)$ and $\int \frac{[5-3x]}{x-2} dx = k \log|x-2| + c$

then $K =$

- 1) 1 2) 2 3) 3 4) 4

52. If $2 < x < 3$ then $\int (|x-1| + |x-2| + |x-3|) dx =$

- 1) $-\frac{x^2}{2} + c$ 2) $x + c$ 3) $\frac{x^2}{2} + c$ 4) $-x + c$

53. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = ax + b \log(9e^{2x} + c) + d$, then descending order of a, b, c, is

- 1) a, b, c 2) b, c, a 3) b, a, c 4) c, a, b

54. $\int (x^{-11} C_1 x^2 + {}^{11}C_2 x^3 - {}^{11}C_3 x^4 + \dots - {}^{11}C_{11} x^{12}) dx$ equals

- 1) $\frac{(1-x)^{12}}{12} - \frac{(1-x)^{11}}{11} + c$ 2) $\frac{(1-x)^{13}}{13} - \frac{(1-x)^{12}}{12} + c$
3) $\frac{(1-x)^{11}}{11} - \frac{(1-x)^{12}}{12} + c$ 4) $\frac{(1-x)^{12}}{12} - \frac{(1-x)^{13}}{13} + c$

55. $\int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx$

- 1) $\tan^{-1} x - \tan x + c$ 2) $\tan x - \tan^{-1} x + c$
3) $\tan^{-1} x + \tan x + c$ 4) $-\tan x - \tan^{-1} x + c$

56. If $I_n = \int \frac{\sin^2 nx}{\sin^2 x} dx$ then $I_{n+1} - 2I_n + I_{n-1} =$

- 1) $\frac{1}{n} \sin 2nx + c$ 2) $-\frac{1}{n} \sin 2nx + c$
3) $\frac{1}{n} \cos 2nx + c$ 4) $-\frac{1}{n} \cos 2nx + c$

57. $\int \frac{1}{\tan^2 x + \sec^2 x} dx =$

- 1) $\frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) - \tan^{-1}(\tan x) + c$
2) $\frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) - \tan^{-1}(\tan x) + c$
3) $\sqrt{2} \tan^{-1}(\sqrt{2} \tan x) - \tan^{-1}(\tan x) + c$
4) $\sqrt{2} \tan^{-1}(\tan x) - \tan^{-1}(\sqrt{2} \tan x) + c$

58. The value of constant of integration which makes the value of integral of $\sin 3x \cdot \cos 5x$ is equal to zero at $x = 0$ is

- 1) $\frac{1}{8}$ 2) $\frac{3}{16}$ 3) $\frac{-3}{16}$ 4) $\frac{-5}{6}$

59. $\int \frac{\sqrt{x^2 + 1}}{x^4} dx$ is equal to

- 1) $-\frac{1}{3} \frac{(x^2 + 1)^{\frac{3}{2}}}{x^3} + e$ 2) $x^3 (x^2 + 1)^{\frac{-1}{2}} + c$
3) $\frac{\sqrt{x^2 + 1}}{x^2} + e$ 4) $-\frac{(x^2 + 1)^{\frac{3}{2}}}{3x^2} + e$

60. If $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \log |\sin x - 2 \cos x| + k$ then a is equal to (AIE-2012)

- 1) -1 2) -2 3) 1 4) 2

61. If $\int f(x) dx = \psi(x)$, then $\int x^5 f(x^3) dx$ is equal to (AIE-2013)

- 1) $\frac{1}{3} [x^3 \psi(x^3) + 3 \int x^2 \psi(x^3) dx] + C$
2) $\frac{1}{3} x^3 \psi(x^3) - 3 \int x^3 \psi(x^3) dx + C$
3) $\frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + C$
4) $\frac{1}{3} [x^3 \psi(x^3) - \int x^3 \psi(x^3) dx] + C$

LEVEL-II (C.W.) - KEY

1)2	2)3	3)4	4)2	5)4	6)4
7)2	8)2	9)4	10)1	11)1	12)3
13)1	14)1	15)3	16)1	17)2	18)2
19)2	20)1	21)2	22)1	23)1	24)3
25)4	26)3	27)1	28)4	29)1	30)3
31)2	32)3	33)2	34)4	35)3	36)4
37)3	38)2	39)1	40)2	41)2	42)3
43)3	44)4	45)1	46)1	47)2	48)3
49)3	50)4	51)2	52)3	53)3	54)2
55)2	56)1	57)3	58)3	59)1	60)4
61)3					

LEVEL-II (C.W.) - HINTS

- $I = \int \left[(\sec x + \tan x)^2 \right]^{\frac{1}{2}} dx$
- Put $1 + \frac{1}{x^2} = t^2$ and use byparts
- divide with x^{2n} $G.I = \int \frac{mx^{m-1} - \frac{n}{x^{n+1}}}{\left(x^m + \frac{1}{x^n}\right)^2} dx$ and Put $x^m + \frac{1}{x^n} = t$.
- $\cos 3x = 4 \cos^3 x - 3 \cos x$
- $\int \cot^4 x = \frac{-1}{3} \cot^3 x + \cot x + x + c$
- $\int \frac{\sin^8 x - \cos^8 x}{1 - 2 \sin^2 x \cos^2 x} dx = \int (\sin^4 x - \cos^4 x) dx = -\int \cos 2x dx$
- Put $1 + x^n = t^2$
- we have, $\int (x^x)^x (2x \log_e x + x) dx$
 $= \int x^{x^2} (2x \log_e x + x) dx = x^{x^2} + C = (x^x)^x + C$
- $\sin^{\frac{1}{2}} x \cos^{\frac{7}{2}} x = \cos^4 x \tan^{\frac{1}{2}} x$
 $\Rightarrow I = \int \frac{\sec^4 x}{\sqrt{\tan x}} dx, \text{ put } \tan x = t^2$
- Putting $\log x = t$ and $\frac{1}{x} dx = dt$ we get

$$\int \frac{1}{x \left\{ 6(\log x)^2 + 7 \log x + 2 \right\}} dx$$

$$= \frac{1}{6} \int \frac{1}{(\log x)^2 + \frac{7}{6} \log x + \frac{1}{3}} d(\log x)$$

$$= \log \left| \frac{2 \log x + 1}{3 \log x + 2} \right| + C$$
- Put $x = z^6 \Rightarrow dx = 6z^5 dz$

$$= 6 \int \left(z^5 + z^4 + z^3 + z^2 + z + 1 + \frac{1}{z-1} \right) dz$$

- $$= 6 \left[\frac{x}{6} + \frac{x^{5/6}}{5} + \frac{x^{2/3}}{4} + \frac{x^{1/2}}{3} + \frac{x^{1/3}}{2} + x^{1/6} + \log(x^{1/6} - 1) \right] + C$$
- Put $x + \sqrt{1+x^2} = z; \Rightarrow \left(1 + \frac{x}{\sqrt{1+x^2}} \right) dx = dz$
 $\Rightarrow x = \frac{z^2 - 1}{2z} = \int z^n \cdot \left(\frac{z-x}{z} \right) dz = \int z^n \frac{2z^2 - (z^{n-1})}{2z^2} dz$
 $z = x + \sqrt{1+x^2}$
 - Put $x = \sin^2 \theta. \quad \therefore I = 2 \int \frac{1 - \sin \theta}{\cos^2 \theta} d\theta$
 - Put $x = a \tan \theta$
 - Put $x = \tan \theta$
 - Put $x = z^6 \quad \therefore G.I = 6 \int \left(z^3 + \frac{1}{z^2 + 1} \right) dz.$
 - Put $2 + x^{2/3} = z^4 \quad \therefore G.I = \int x^{1/3} \cdot z (6z^3 \cdot x^{1/3}) dz$
 $= 6 \int z^4 (z^4 - 2) dz$
 - Put $1 + \frac{1}{x^2} = t.$
 - $\frac{x-q}{x-p} = t$
 - Put $t = \frac{x-2}{x+3}$
 - Using by parts
 - use $\int UV$
 - Put $x^2 = t$
 - Let $I = \int f^{-1}(x) dx$ Put $f^{-1}(x) = u$
 $\therefore I = \int u \cdot f^1(u) dx$ and using byparts
 - $U = \log \left(\tan \frac{x}{2} \right), V = \cos x$
 - $GI = x^2 \left(\frac{-1}{x \tan x + 1} \right) - \int \frac{2x(-1)}{x \tan x + 1} dx$
 $= \frac{-x^2}{x \tan x + 1} + 2 \int \frac{x \cos x}{x \sin x + \cos x} dx$

27. Put $\log x = t$

28. $\int e^x (\sin^2 2x + 2 \sin 4x - 2 \sin 4x - 8 \cos 4x) dx$
 $= \int e^x (\sin^2 2x + 2 \sin 4x) - 2 \int e^x (\sin 4x + 4 \cos 4x)$
 $= e^x (\sin^2 2x - 2 \sin 4x) + c$

29. Use the formula $\int e^x [f(x) + f'(x)] dx$

30. Put $\log_e x = t \Rightarrow x = e^t$

31. Put $\log x = t$

32. Put $\log(x^4) = t$

33. Put $-\frac{x}{2} = t$

34. $\int e^x \left\{ \frac{1 + nx^{n-1} - x^{2n}}{(1-x^n)\sqrt{1-x^{2n}}} \right\} dx$

$\int e^x \left[\frac{\sqrt{1-x^{2n}}}{1-x^n} + \frac{nx^{n-1}}{(1-x^n)\sqrt{1-x^{2n}}} \right] = e^x \frac{\sqrt{1-x^{2n}}}{1-x^n} + c$

35. $f(x) = \int e^x (x-1)(x-2) dx$

For decreasing function, $f'(x) < 0$

$\Rightarrow e^x (x-1)(x-2) < 0 \Rightarrow (x-1)(x-2) < 0$

$\Rightarrow 1 < x < 2$ (since $e^x > 0 \forall x \in R$)

36. Apply the formulae $\int e^{ax} \sin bxdx$ and $\int e^{ax} \cos bxdx$

37. Put $-x^2 = t$

38. Put $x + \frac{1}{x} = t$

$\therefore I = \int \frac{-dt}{t\sqrt{t^2-2}} = \int \frac{-dt}{t^2 \sqrt{1 - \left(\frac{\sqrt{2}}{t}\right)^2}}$

39. $\sin 2\theta = (\sin \theta + \cos \theta)^2 - 1$

40. $\int \frac{1}{(a^2 - x^2)(a^2 + x^2)} dx$
 $= \frac{1}{2a^2} \int \frac{(a^2 + x^2) + (a^2 - x^2)}{(a^2 + x^2)(a^2 - x^2)} dx$

41. $I = \int \left[1 - \frac{x+1}{x(x^2+1)} \right] dx$

$I = \int \left[1 - \frac{1}{x^2+1} - \left(\frac{1}{x} - \frac{x}{x^2+1} \right) \right] dx$

42. Put $x^6 = t$ and using partial fractions

43. Put $x+1 = \frac{1}{t}$

44. $U = x^m, V = \frac{1}{(\log x)^n}$

45. Multiply Numerator & Denominator by $\sin(b-a)$

46. $1 + \sin x = 1 + \cos\left(\frac{\pi}{2} - x\right)$

47. $I = \int \frac{x^3 \left(x - \frac{1}{x^3}\right)}{x^3 \sqrt{x^2 + 1 + \frac{1}{x^2}}} dx$ and put $\left(x^2 + 1 + \frac{1}{x^2}\right) = t$

48. Let $f(x) = x^n + 1 \therefore 82 = 3^n + 1 \Rightarrow n = 4$

49. $I = \int \frac{(x^2 \sec^2 x + \tan^2 x)}{1+x^2} dx$
 $= \int \frac{(x^2+1) \sec^2 x - 1}{(x^2+1)} dx$ and split it.

50. $I = \frac{1}{\sqrt{2}} \int \frac{dx}{(\sin x \cdot \sin \pi/4 - \cos x \cdot \cos \pi/4 + 1)}$

$I = \frac{1}{\sqrt{2}} \int \frac{dx}{1 - \cos\left(x + \frac{\pi}{4}\right)} = \frac{1}{2\sqrt{2}} \int \sec^2 \left(\frac{x}{2} + \frac{\pi}{8}\right) dx$

51. $\frac{2}{3} < x < 1 ; 2 < 5-3x < 3 \Rightarrow [5-3x] = 2$

52. $2 < x < 3 \Rightarrow x-1 > 0, x-2 > 0, x-3 < 0$

53. Use $Nr = A(Dr) + B \frac{d}{dx}(Dr)$

54. $\int (x^{-11} C_1 x^2 + {}^{11}C_2 x^3 - {}^{11}C_3 x^4 \dots - {}^{11}C_{11} x^{12}) dx$
 $= \int x(1-x)^{11} dx$ put $1-x = t$

then, $I = -\int (1-t)t^{11} dt = -\frac{t^{12}}{12} + \frac{t^{13}}{13} + c$

$= \frac{(1-x)^{13}}{13} - \frac{(1-x)^{12}}{12} + c$

$$55. \int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx =$$

$$\int \frac{(1 + x^2 - 1 + \sin^2 x)}{1 + x^2} \sec^2 x dx$$

$$\int \frac{(1 + x^2) \sec^2 x - \sec^2 x + \tan^2 x}{(1 + x^2)} dx$$

$$\int \sec^2 x dx - \int \frac{1}{1 + x^2} dx = \tan x - \tan^{-1} x + c$$

$$56. I_{n+1} - 2I_n + I_{n-1} = (I_{n+1} - I_n) - (I_n - I_{n-1})$$

$$57. \int \frac{\sec^2 x}{(\tan^2 x + \sec^2 x) \sec^2 x} dx =$$

$$\int \frac{\sec^2 x}{(1 + 2 \tan^2 x)(1 + \tan^2 x)} dx$$

$$\text{put } \sqrt{2} \tan x = t$$

$$58. y = \int \sin 3x \cdot \cos 5x dx; \frac{1}{2} (\sin 8x - \sin 2x) dx$$

$$y = \frac{1}{2} \left[\frac{\cos 2x}{2} - \frac{\cos 8x}{8} \right] + c \rightarrow (1)$$

$$\text{take } x = 0, y = 0$$

$$59. x = \tan \theta \Rightarrow I = \int \frac{\cos \theta}{\sin^4 \theta} d\theta$$

$$60. \int \frac{5 \sin x}{\sin x - 2 \cos x} dx = x + 2 \log |\sin x - 2 \cos x| + k$$

$$61. \int x^5 f(x^3) dx \Rightarrow I = \frac{1}{3} \int 3x^2 \cdot x^3 f(x^3) dx$$

$$\text{put } x^3 = t \therefore I = \frac{1}{3} \int t \cdot f(t) dt$$

$$I = \frac{1}{3} \left[t \int f(t) dt - \int (f(t) dt) t \right]$$

$$\therefore I = \frac{1}{3} \left[x^3 \psi(x^3) - \int \psi(t) dt \right]$$

$$I = \frac{1}{3} \left[x^3 \cdot \psi(x^3) - \int \psi(x^3) \cdot 3x^2 dx \right]$$



LEVEL-II (H.W)

INTEGRATION BY THE METHOD OF SUBSTITUTION

$$1. \text{ If } \int \frac{\cos^4 x}{\sin^2 x} dx = A \cot x + B \sin 2x + C \frac{x}{2} + D, \text{ then}$$

$$1) B = -2$$

$$2) B = -3/2$$

$$3) B = -1/4$$

$$4) B = -1$$

$$2. \int \frac{x^5}{\sqrt{1+x^3}} dx =$$

$$1) \frac{2}{9} (x^3 - 2) \sqrt{1+x^3} + c$$

$$2) \frac{2}{9} (x^3 + 2) \sqrt{1+x^3} + c$$

$$3) \frac{1}{9} (x^3 - 1) \sqrt{1+x^3} + c \quad 4) \frac{1}{9} (x^3 + 1) \sqrt{1+x^3} + c$$

$$3. \int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx =$$

$$1) \tan^{-1}(\tan^2 x) + c$$

$$2) \tan^{-1}(\cos^2 x) + c$$

$$3) \tan^{-1}(\sin^2 x) + c$$

$$4) \tan^{-1}(\cot^2 x) + c$$

$$4. \int \frac{x}{x - \sqrt{x^2 - 1}} dx \text{ is equal to}$$

$$1) \frac{x^3}{3} + \frac{1}{2} (x^2 - x - 1)^{\frac{3}{2}} + c$$

$$2) \frac{x^3}{3} + \frac{1}{3} (x^2 - 1)^{\frac{3}{2}} + c$$

$$3) \frac{x^3}{3} + \frac{2}{3} (x^2 - 1)^{\frac{3}{2}} + c \quad 4) \frac{x^3}{3} + (x^2 - 1)^{\frac{3}{2}} + c$$

$$5. \int \frac{(x^2 - 1)}{(x^4 + 3x^2 + 1) \tan^{-1} \left(\frac{x^2 + 1}{x} \right)} dx =$$

$$1) \log \left| \tan^{-1} \left(\frac{x-1}{x} \right) \right| + c \quad 2) \log \left| \tan^{-1} \left(x + \frac{1}{x} \right) \right| + c$$

$$3) \log \left| \tan^{-1} (x^2 + 1) \right| + c \quad 4) \log \left| \tan^{-1} \left(\frac{x+1}{x} \right) \right| + c$$

6. If $I^r(x)$ means $\log \log \log \dots \log x$, the log being repeated 'r' times, then

$$\int \{x I(x) I^2(x) I^3(x) \dots I^r(x)\}^{-1} dx =$$

- 1) $I^{r+1}(x) + c$ 2) $\frac{I^{r+1}(x)}{r+1} + c$
3) $I^r(x) + c$ 4) $\frac{I^r(x)}{r} + c$

7. $\int \sqrt{\frac{\cos x - \cos^3 x}{1 - \cos^3 x}} dx =$

- 1) $\frac{2}{3} \sin^{-1} \left(\cos^{\frac{3}{2}} x \right) + c$ 2) $\frac{3}{2} \sin^{-1} \left(\cos^{\frac{3}{2}} x \right) + c$
3) $\frac{2}{3} \cos^{-1} \left(\cos^{\frac{3}{2}} x \right) + c$ 4) $\frac{3}{2} \cos^{-1} \left(\cos^{\frac{3}{2}} x \right) + c$

8. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx =$

- 1) $\tan x + \cot x + c$ 2) $\tan x + \cot x + 3x + c$
3) $\tan x - \cot x - 3x + c$ 4) $\tan x + \cot x - 3x + c$

9. $\int \frac{dx}{\cos^4 x \sin^2 x} =$

- 1) $\frac{1}{2} \tan^3 x + \tan x - \cot x + C$
2) $\frac{1}{3} \tan^3 x + 2 \tan x + \cot x + c$
3) $\frac{1}{3} \tan^3 x + 2 \tan x - \cot x + c$
4) $\frac{1}{3} \tan^3 x - \tan x - \cot x + c$

10. $\int \frac{\cos 2x - \sin 2x}{\cos 2x + \sin 2x} dx =$

- 1) $-\frac{1}{2} \log \left| \cos \left(\frac{\pi}{4} - 2x \right) \right| + c$
2) $\frac{1}{2} \log \left| \cos \left(\frac{\pi}{4} - 2x \right) \right| + c$
3) $\tan 2x + c$
4) $\cot 2x + c$

11. $\int \cos^{-3/7} x \sin^{-11/7} x dx$ is equal to

- 1) $\log |\sin^{4/7} x| + c$ 2) $\frac{4}{7} \tan^{4/7} x + c$
3) $-\frac{7}{4} \tan^{-4/7} x + c$ 4) $\log |\cos^{3/7} x| + c$

12. The value of the integral $\int \frac{x \sin x^2 e^{\sec x^2}}{\cos^2 x^2} dx$ is

- 1) $\frac{1}{2} e^{\sec x^2} + C$ 2) $\frac{1}{2} e^{\sin x^2} + C$
3) $\frac{1}{2} e^{\sin x^2} e^{\cos x^2} + C$ 4) $e^{\sec x^2} + c$

SUBSTITUTION OF TRIGONOMETRIC FUNCTIONS

13. $\int \frac{1}{(a^2 + x^2)^{3/2}} dx =$

- 1) $\frac{1}{a^2} \cdot \frac{x}{\sqrt{a^2 + x^2}} + c$
2) $-\frac{1}{a^2} \cdot \frac{x}{\sqrt{a^2 + x^2}} + c$
3) $\frac{1}{a^2} \cdot \frac{x}{(a^2 + x^2)^{3/2}} + c$
4) $\frac{-1}{a^2} \cdot \frac{x}{(a^2 + x^2)^{3/2}} + c$

14. $\int \frac{dt}{t + \sqrt{a^2 - t^2}}$ equal to

- 1) $\frac{1}{2} \sin^{-1} \left(\frac{t}{a} \right) + \log \left(t + \sqrt{a^2 - t^2} \right) + k$
2) $\frac{1}{2} \sin^{-1} \left(\frac{t}{a} \right) + \log \sqrt{t + \sqrt{a^2 - t^2}} + k$
3) $\frac{1}{2} \sin^{-1} \left(\frac{t}{a} \right) + \log \sqrt{a + \sqrt{a^2 - t^2}} + k$
4) $\frac{1}{2} \sin^{-1} \left(\frac{t}{a} \right) + \log \left(a + \sqrt{a^2 - t^2} \right) + k$

15. $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}} =$

1) $\frac{-1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$

2) $\frac{-1}{\sqrt{2}} \cot^{-1} \left(\frac{\sqrt{2x}}{\sqrt{1-x^2}} \right) + c$

3) $-\frac{1}{2} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$

4) $\frac{1}{2} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$

RATIONAL INDEX

16. If $\int \frac{dx}{\sqrt[4]{x^3}(\sqrt[4]{x}+1)^{10}} = k \cdot \frac{1}{(\sqrt[4]{x}+1)^9} + c$, then

1) $k = 4/9$ 2) $k = -9/4$ 3) $k = -4/9$ 4) $k = 9/4$

17. $\int \frac{\sqrt[3]{1+\sqrt[4]{x}}}{\sqrt{x}} dx$ is equal to

1) $12 \left[\frac{(1+\sqrt[4]{x})^{7/3}}{7} + \frac{(1+\sqrt[4]{x})^{4/3}}{4} \right] + c$

2) $12 \left[\frac{(1+\sqrt[4]{x})^{7/3}}{7} - \frac{(1+\sqrt[4]{x})^{4/3}}{4} \right] + c$

3) $\frac{1}{12} \left[\left(\frac{1+x^{1/4}}{4} \right)^{4/3} - x \right] + c$

4) $\frac{1}{12} \left[\left(\frac{1+x^{1/4}}{4} \right)^{4/3} + x \right] + c$

18. $\int \frac{2x}{(1-x^2)\sqrt{x^4-1}} dx =$

1) $\sqrt{\frac{x^2+1}{x^2-1}} + c$ 2) $\sqrt{\frac{x^2-1}{x^2+1}} + c$

3) $\sqrt{x^4-1} + c$ 4) $\sqrt{x^4+1} + c$

19. $\int \frac{1}{(x+1)^{3/2}(x-1)^{1/2}} dx =$

1) $-\sin^{-1} \left[\frac{1-x}{(x+2)\sqrt{2}} \right] + c$ 2) $\sqrt{\frac{x+1}{x-1}} + c$

3) $\sqrt{1-\frac{2+x}{x+1}} + c$ 4) $\sqrt{\frac{x-1}{x+1}} + c$

INTEGRATION BY PARTS

20. $\int x \frac{(\sec 2x - 1)}{(\sec 2x + 1)} dx =$

1) $x \tan x + \log |\sec x| - \frac{x^2}{2} + c$

2) $x \tan x - \log |\sec x| - \frac{x^2}{2} + c$

3) $x \tan x - \log |\sec \frac{x}{2}| + c$

4) $x \tan x - \log |\sec \frac{x}{2}| + \frac{x^2}{2} + c$

21. $\int \frac{\arcsin \sqrt{x}}{\sqrt{1-x}} dx =$

1) $2[\sqrt{x} - \sqrt{1-x} \arcsin \sqrt{x}] + c$

2) $2[\sqrt{x} + \sqrt{1-x} \arcsin \sqrt{x}] + c$

3) $2[\sqrt{x} + \sqrt{1-x} \arcsin \sqrt{x}] + c$

4) $2[\sqrt{x} - \sqrt{1-x} \arcsin \sqrt{x}] + c$

22. $\int \sin x \log (\sec x + \tan x) dx = f(x) + x + c$

then $f(x) =$

1) $\cos x \log (\sec x + \tan x) + c$

2) $\sin x \log (\sec x + \tan x) + c$

3) $-\cos x \log (\sec x + \tan x) + c$

4) $-\cos x \log \sec x + c$

23. The value of

$\int e^{\sec x} \cdot \sec^3 x (\sin^2 x + \cos x + \sin x + \sin x \cos x) dx$ is

1) $e^{\sec x} (\sec^2 x + \sec x \tan x) + C$

2) $e^{\sec x} + C$

3) $e^{\sec x} (\sec x + \tan x) + C$

4) $e^{\sec x} (\sec x - \tan x) + C$

24. The value of $\int \frac{e^x(2-x^2)dx}{(1-x)\sqrt{1-x^2}}$ is equal to

- 1) $e^x \sqrt{\frac{1+x}{1-x}} + c$ 2) $e^x \sqrt{1+x} + c$
 3) $e^x \sqrt{1-x} + c$ 4) $e^x \sqrt{\frac{1-x}{1+x}} + c$

25. $\int \left(\frac{3-x^2}{1-2x+x^2} \right) e^x dx = e^x f(x) + c \Rightarrow f(x) =$

- 1) $\frac{1+x}{1-x}$ 2) $\frac{1-x}{1+x}$
 3) $\frac{1+x}{x-1}$ 4) $\frac{x-1}{1+x}$

III. Problems based on $\int e^{ax} \sin bx \, dx$ and

$$\int e^{ax} \cos bx \, dx$$

26. $\int \frac{\sin(\log x)}{x^3} dx =$

- 1) $\frac{-1}{5x^2} (2 \sin(\log x) + \cos(\log x)) + c$
 2) $\frac{1}{5x^2} (2 \sin(\log x) + \cos(\log x)) + c$
 3) $\frac{1}{5x^2} (2 \sin(\log x) - \cos(\log x)) + c$
 4) $\frac{1}{x^2} (2 \sin(\log x) - \cos(\log x)) + c$

27. If $\int \sin(\log x) dx = \frac{x}{\sqrt{2}} f(x) + c$, then

- 1) $f(x) = \sin\left(\log x - \frac{\pi}{4}\right)$
 2) $f(x) = \cos\left\{\log x + \frac{\pi}{4}\right\}$
 3) $f(x) = \sin(\log x) - \cos(\log x)$
 4) $f(x) = \sin\left(\frac{\log x}{2}\right)$

PROBLEMS ON INVERSE TRIGONOMETRIC FUNCTIONS

28. $\int \cos^{-1} \sqrt{\frac{1-x}{2}} dx =$

- 1) $\frac{\pi}{2} x - \frac{1}{2} x \cos^{-1} x + c$
 2) $\frac{\pi}{2} x - \frac{1}{2} x \cos^{-1} x + \frac{1}{2} \sqrt{1-x^2} + c$
 3) $\frac{\pi}{2} x - \frac{1}{2} x \cos^{-1} x - \sqrt{1-x^2} + c$
 4) $\frac{\pi}{2} x + \frac{1}{2} x \cos^{-1} x + c$

29. $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx =$

- 1) $\frac{x + \tan^{-1} x}{(1+x^2)^{3/2}} + c$ 2) $\frac{x - \tan^{-1} x}{\sqrt{(1+x^2)}} + c$
 3) $\frac{x}{\sqrt{1+x^2}} - \tan^{-1} x + c$ 4) $\frac{x}{1+x^2} + \tan^{-1} x + c$

30. $\int \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right) dx =$

- 1) $\frac{1}{2} \left[x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right] + c$
 2) $x \tan^{-1} x - \frac{1}{2} \log(1+x^2) + c$
 3) $x \tan^{-1} x + \frac{1}{2} \log(1+x^2) + c$
 4) $\frac{1}{2} [x \tan^{-1} x - \log(1+x^2)] + c$

APPLICATIONS OF STANDARD INTEGRALS

31. If $\int \frac{x^2+4}{x^4+16} dx = \frac{1}{k} \tan^{-1} \left(\frac{x^2-4}{kx} \right) + C$ then $k =$

- 1) $\sqrt{2}$ 2) $4\sqrt{2}$
 3) $2\sqrt{2}$ 4) 2

32. $\int \frac{e^{3x} + e^x}{e^{4x} - e^{2x} + 1} dx =$

- 1) $\frac{1}{4} \log(e^{4x} - e^{2x} + 1) + c$ 2) $\tan^{-1}(e^x - e^{-x}) + c$
 3) $\tan^{-1}(e^x + e^{-x}) + c$ 4) $\tan^{-1}(e^{-x} - e^x) + c$

33. If $\int \frac{dx}{x^4 + 5x^2 + 4} = A \tan^{-1} x + B \tan^{-1} \left(\frac{x}{2} \right) +$

C, then B =

- 1) 1/6 2) -1/6 3) 6 4) -6

34. $\int \frac{x^2 + 2}{(x^2 + 1)(x^2 + 4)} dx = A \tan^{-1} x + B \tan^{-1} \frac{x}{2} + c$

then (A, B) =

- 1) $\left(\frac{1}{4}, \frac{1}{3} \right)$ 2) $\left(\frac{1}{2}, \frac{1}{3} \right)$ 3) $\left(\frac{-1}{2}, \frac{-2}{3} \right)$ 4) $\left(\frac{1}{3}, \frac{1}{3} \right)$

35. If $\int \frac{1}{x(x^5 - 1)(x^5 + 1)} dx =$

$A \log x + B \log(x^5 - 1) + C \log(x^5 + 1)$ then

(A, B, C) =

- 1) $\left(-1, \frac{1}{10}, \frac{1}{10} \right)$ 2) $\left(-\frac{5}{4}, \frac{1}{8}, -\frac{1}{8} \right)$
 3) $\left(-\frac{5}{4}, -\frac{1}{8}, \frac{1}{8} \right)$ 4) $\left(\frac{1}{5}, \frac{1}{10}, -\frac{1}{10} \right)$

36. If $\int \frac{6x+7}{(x+2)^2} dx = A \log|x+2| + \frac{B}{x+2} + c$ then

(A, B) =

- 1) $\left(6, \frac{1}{5} \right)$ 2) $(-6, -5)$ 3) $\left(\frac{1}{6}, 5 \right)$ 4) $(6, 5)$

37. $\int \frac{dx}{1 + \sqrt[3]{x+1}} =$

- 1) $\frac{3}{2}(x+1)^{2/3} - 3(x+1)^{1/3} + 3 \log|1 + \sqrt[3]{x+1}| + c$
 2) $\frac{3}{2}(x+1)^{2/3} + 3(x+1)^{1/3} + 3 \log|1 + \sqrt[3]{x+1}| + c$
 3) $\frac{3}{2}(x+1)^{2/3} + 3(x+1)^{1/3} + c$
 4) $(x+1)^{2/3} + 3(x+1)^{1/3} + c$

38. $\int \sqrt{\frac{x-1}{2x-3}} dx =$

- 1) $\sin^{-1} x + \sqrt{1-x^2} + c$
 2) $\cos^{-1} x - \sqrt{1-x^2} + c$
 3) $\sin^{-1} x - \sqrt{1-x^2} + c$
 4) $\frac{1}{2} \sqrt{2x^2 - 5x + 3} + \frac{1}{4\sqrt{2}} \cosh^{-1}(4x-5) + c$

REDUCTION FORMULAE

39. If $I_n = \int \tan^n x dx,$

then $I_0 + I_1 + 2(I_2 + \dots + I_8) + I_9 + I_{10} =$

- 1) $\frac{\tan x}{1} + \frac{\tan^2 x}{2} + \dots + \frac{\tan^9 x}{9}$
 2) $-\left(\frac{\tan x}{1} + \frac{\tan^2 x}{2} + \dots + \frac{\tan^9 x}{9} \right)$
 3) $\frac{\cot x}{1} + \frac{\cot^2 x}{2} + \dots + \frac{\cot^9 x}{9}$
 4) $-\left(\frac{\cot x}{1} + \frac{\cot^2 x}{2} + \dots + \frac{\cot^9 x}{9} \right)$

40. If $I_{m,n} = \int x^m (\log x)^n dx$

$= \frac{x^{m+1}}{m+1} (\log x)^n + f(x),$ then $f(x) =$

- 1) $\frac{n}{n+1} I_{m,n-1}$ 2) $\frac{-m}{n+1} I_{m,n-1}$
 3) $\frac{m}{n+1} I_{m,n-1}$ 4) $\frac{-n}{m+1} I_{m,n-1}$

41. If $I_n = \int (\log x)^n dx,$ then $I_6 + 6I_5 =$

- 1) $x(\log x)^5$ 2) $-x(\log x)^5$
 3) $x(\log x)^6$ 4) $-x(\log x)^6$

42. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + c,$

then value of (A, B) is

- 1) $(\sin \alpha, \cos \alpha)$ 2) $(-\cos \alpha, \sin \alpha)$
 3) $(-\sin \alpha, \cos \alpha)$ 4) $(\cos \alpha, \sin \alpha)$

43. If $\int (\sin 2x - \cos 2x) dx = \frac{1}{\sqrt{2}} \sin(2x-a) + c$

then

1) $a = \frac{7\pi}{4}$ 2) $a = -\frac{5\pi}{4}$

3) $a = -\frac{\pi}{4}$ 4) $a = \frac{-3\pi}{4}$

44. $\int \frac{1}{x^2} \sqrt{\frac{x-1}{x+1}} dx =$

1) $\cos^{-1} \frac{1}{|x|} - \frac{\sqrt{x^2-1}}{x} + c$

2) $\cos^{-1} \frac{1}{|x|} + \frac{\sqrt{x^2-1}}{x} + c$

3) $\sin^{-1} \frac{1}{|x|} - \frac{\sqrt{x^2-1}}{x} + c$

4) $\sin^{-1} \frac{1}{|x|} + \frac{\sqrt{x^2-1}}{x} + c$

45. $\int \sqrt{x-3} (\sin^{-1}(\ln x) + \cos^{-1}(\ln x)) dx$ equal to

1) $\frac{\pi}{3}(x-3)^{3/2} + k$ 2) $\frac{\pi}{2}(x-3)^{3/2} + k$

3) $\frac{\pi}{2}x^{3/2} + k$ 4) Does not exist

46. $\int \frac{7x^8 + 8x^7}{(1+x+x^8)^2} dx = f(x) + c \Rightarrow f(x) =$

1) $\frac{x^8}{1+x+x^8}$ 2) $28 \log(1+x+x^8)$

3) $\frac{1}{1+x+x^8}$ 4) $\frac{-1}{1+x+x^8}$

47. If $\int \frac{1}{\sin x + \cos x} dx = k \log \left| \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) \right| + c$, then $k =$

1) $\sqrt{2}$ 2) $\frac{1}{\sqrt{2}}$ 3) $-\sqrt{2}$ 4) $\frac{-1}{\sqrt{2}}$

48. $\int \frac{dx}{\cos x - \sin x} =$

1) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + c$

2) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + c$

3) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + c$

4) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + c$

49. $\int \frac{1}{\sqrt{1-\sin x}} dx = \text{for } x \in \left(\frac{\pi}{4}, \frac{\pi}{2} \right)$

1) $\sqrt{2} \tan \left(\frac{3\pi}{8} + \frac{x}{4} \right) + c$ 2) $\sqrt{2} \cot \left(\frac{3\pi}{8} + \frac{x}{4} \right) + c$

3) $\sqrt{2} \log \left| \tan \left(\frac{3\pi}{8} + \frac{x}{4} \right) \right| + c$

4) $\sqrt{2} \log \left| \cot \left(\frac{3\pi}{8} + \frac{x}{4} \right) \right| + c$

50. $\int \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} dx$

1) $\frac{x^2}{2} + x + c$ 2) $x - \frac{x^2}{2} + c$

3) $\frac{x^2}{2} - x + c$ 4) $\frac{x}{2} - x^2 + c$

51. $\int (x+1)(x+2)^7(x+3) dx$

1) $\frac{(x+2)^{10}}{10} - \frac{(x+2)^8}{8} + c$

2) $\frac{(x+1)^2}{2} - \frac{(x+2)^8}{8} - \frac{(x+3)^2}{2} + c$

3) $\frac{(x+1)^2}{2} + \frac{(x+2)^8}{8} - \frac{(x+3)^2}{2} + c$

4) $\frac{(x+2)^9}{9} - \frac{(x+2)^{10}}{10} + c$

LEVEL-II (H.W.) - KEY

1)3	2)1	3)1	4)2	5)2	6)1
7)3	8)3	9)3	10)2	11)3	12)1
13)1	14)2	15)1	16)3	17)2	18)1
19)4	20)2	21)1	22)3	23)3	24)1
25)1	26)1	27)1	28)2	29)2	30)1
31)3	32)2	33)2	34)4	35)1	36)4
37)1	38)4	39)1	40)4	41)3	42)4
43)2	44)1	45)4	46)1	47)2	48)2
49)3	50)3	51)1			

LEVEL-II (H.W.) - HINTS

- $\cos^4 x = (1 - \sin^2 x)^2$
- Put $1 + x^3 = t^2$
- $\frac{\sin 2x}{\sin^4 x + \cos^4 x} = \frac{2 \tan x \sec^2 x}{1 + \tan^4 x}$
- Rationalizing Dr $GI = \int x(x + \sqrt{x^2 - 1}) dx$
 $= \int x^2 dx + \frac{1}{2} \int (x^2 - 1)^{\frac{1}{2}} 2x dx$
- Put $\tan^{-1}\left(\frac{x^2 + 1}{x}\right) = t$
- Put $I'(x) = t$
- Put $\cos^{\frac{3}{2}} x = t$
- $\sin^6 x + \cos^6 x = \sin^4 x - \sin^2 x \cos^2 x + \cos^4 x$
- $I = \int \frac{(\sin^2 x + \cos^2 x)^2}{\cos^4 x \sin^2 x} dx$
- $\frac{\cos 2x - \sin 2x}{\cos 2x + \sin 2x} = \frac{1 - \tan 2x}{1 + \tan 2x} = \tan\left(\frac{\pi}{4} - 2x\right)$
- $I = \int \tan^{\frac{11}{7}} x \cdot \sec^2 x dx$ Put $\tan x = t$
- we have, $\int \frac{x \sin x^2 e^{\sec x^2}}{\cos^2 x^2} dx = \frac{1}{2} \int e^{\sec x^2} d(\sec x^2)$
 $= \frac{1}{2} e^{\sec x^2} + C$
- Put $x = a \tan \theta$
- Put $t = a \sin \theta$
 $\therefore I = \frac{1}{2} \int \frac{(\cos \theta + \sin \theta) + (\cos \theta - \sin \theta)}{(\sin \theta + \cos \theta)} d\theta$

- Put $x = \cos \theta$
- Put $\frac{1}{x^4} + 1 = t$
- Put $1 + x^{1/4} = z^3$
 $\therefore I = 12 \int z^3 \cdot x^{1/4} \cdot dz = 12 \int z^3 (z^3 - 1) dz$
- Put $\sqrt{\frac{x^2 + 1}{x^2 - 1}} = t$
- Write $\frac{1}{(x+1)^{3/2}(x-1)^{1/2}} = \frac{1}{\left(\frac{x+1}{x-1}\right)^{3/2}(x-1)^2}$
and let $\frac{x+1}{x-1} = t$
- $x \left(\frac{\sec 2x - 1}{\sec 2x + 1} \right) = x \left(\frac{1 - \cos 2x}{1 + \cos 2x} \right)$
- Put $\sin^{-1} \sqrt{x} = t$
- $U = \log(\sec x + \tan x), V = \sin x$
- $\int e^{\sec x} \cdot \sec^3 x (\sin^2 x + \cos x + \sin x + \sin x \cos x) dx$
 $= \int e^{\sec x} \left[\frac{\sec x \cdot \tan x (\sec x + \tan x) +}{(\sec^2 x + \sec x \cdot \tan x)} \right] dx$
 $= (\sec x + \tan x) e^{\sec x} - \int (\sec x \tan x + \sec^2 x) e^{\sec x} dx$
 $+ \int e^{\sec x} (\sec x \tan x + \sec^2 x) dx$
 $= e^{\sec x} (\sec x + \tan x) + C$
- $I = \int e^x \left[\frac{1 + (1 - x^2)}{(1 - x)\sqrt{1 - x^2}} \right] dx$ and using the formula
 $\int e^x (f(x) + f'(x)) dx$
- $x^2 - 2x + 1 = (1 - x)^2$ and $3 - x^2 = 2 + (1 - x^2)$
- Put $t = \log x$ & $x = e^t$
- Let $I = \int \sin(\log x) dx$ Put $\log x = t$
 $\Rightarrow \frac{1}{x} dx = dt \Rightarrow dx = x dt \Rightarrow dx = e^t dt$
 $= e^t \sin t - \left[\cos t \cdot e^t - \int (-\sin t) e^t dt \right]$

$$\begin{aligned}\therefore 1 &= \frac{1}{2} e^t (\sin t - \cos t) + c = \\ \frac{x}{2} [\sin(\log x) - \cos(\log x)] + c \\ &= \frac{x}{\sqrt{2}} \sin\left(\log x - \frac{\pi}{4}\right) + c \\ \therefore f(x) &= \sin\left(\log x - \frac{\pi}{4}\right)\end{aligned}$$

28. Put $x = \cos \theta$

29. Put $x = \tan \theta$

30. Put $x = \tan \theta$

31. Put $x - \frac{4}{x} = t$

32. Take e^{2x} common

33. Resolve into Partial Fractions

34. $\frac{1}{3} \int \frac{(x^2 + 4) + (2x^2 + 2)}{(x^2 + 1)(x^2 + 4)} dx$

35. $I = \int \frac{1}{x(x^{10} - 1)} dx$

36. put $x + 2 = y$

37. Put $x + 1 = t^3$

38. $\sqrt{\frac{x-1}{2x-3}} = \frac{x-1}{\sqrt{(2x-3)(x-1)}} = \frac{1}{4} \frac{(4x-5)+1}{\sqrt{2x^2-5x+3}}$

39. $I_n + I_{n-2} = \frac{1}{n-1}$

40. Put $u = (\log x)^n$, $V = x^m$

41. Apply the formula $I_n = x(\log x)^n - n.I_{n-1}$

42. $\sin x = \sin[(x - \alpha) + \alpha]$

43. Multiplying and dividing with $\sqrt{2}$

44. Put $\frac{1}{x} = \cos \theta$

45. Domain of $\sqrt{x-3}$ is $x > 3$

then $\log_e x > 1$

But Domain of $\sin^{-1} x$ & $\cos^{-1} x$ is $[-1, 1]$

46.

$$\begin{aligned}G.I &= \int \frac{(7x^8 + 8x^7)}{x^{16} \left(\frac{1}{x^8} + \frac{1}{x^7} + 1 \right)^2} dx \\ &= - \int \frac{\left(\frac{-7}{x^8} - \frac{8}{x^9} \right)}{\left(\frac{1}{x^8} + \frac{1}{x^7} + 1 \right)^2} dx\end{aligned}$$

47. $\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$

48. $\cos x - \sin x = \sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$

49. $\sqrt{1 - \sin x} = \cos \frac{x}{2} - \sin \frac{x}{2} = \sqrt{2} \cos\left(\frac{\pi}{4} + \frac{x}{2}\right)$

50.

$$\begin{aligned}&\int \frac{(\sqrt{x} + 1)\sqrt{x} \left((\sqrt{x})^3 - 1 \right)}{\sqrt{x}(x + \sqrt{x} + 1)} \\ &= \int \frac{(\sqrt{x} + 1)(\sqrt{x} - 1)(x + \sqrt{x} + 1)}{x + \sqrt{x} + 1} \\ &= \int (x - 1) dx = \frac{x^2}{2} - x + c\end{aligned}$$

51. $\int (x + 2 - 1)(x + 2)^7 (x + 2 + 1) dx$

$$\int (t - 1)(t^7)(t + 1) dt$$

$$\int (t^2 - 1)t^7 dt = \int (t^9 - t^7) dt$$

$$= \frac{t^{10}}{10} - \frac{t^8}{8} + c$$

where $x + 2 = t$